

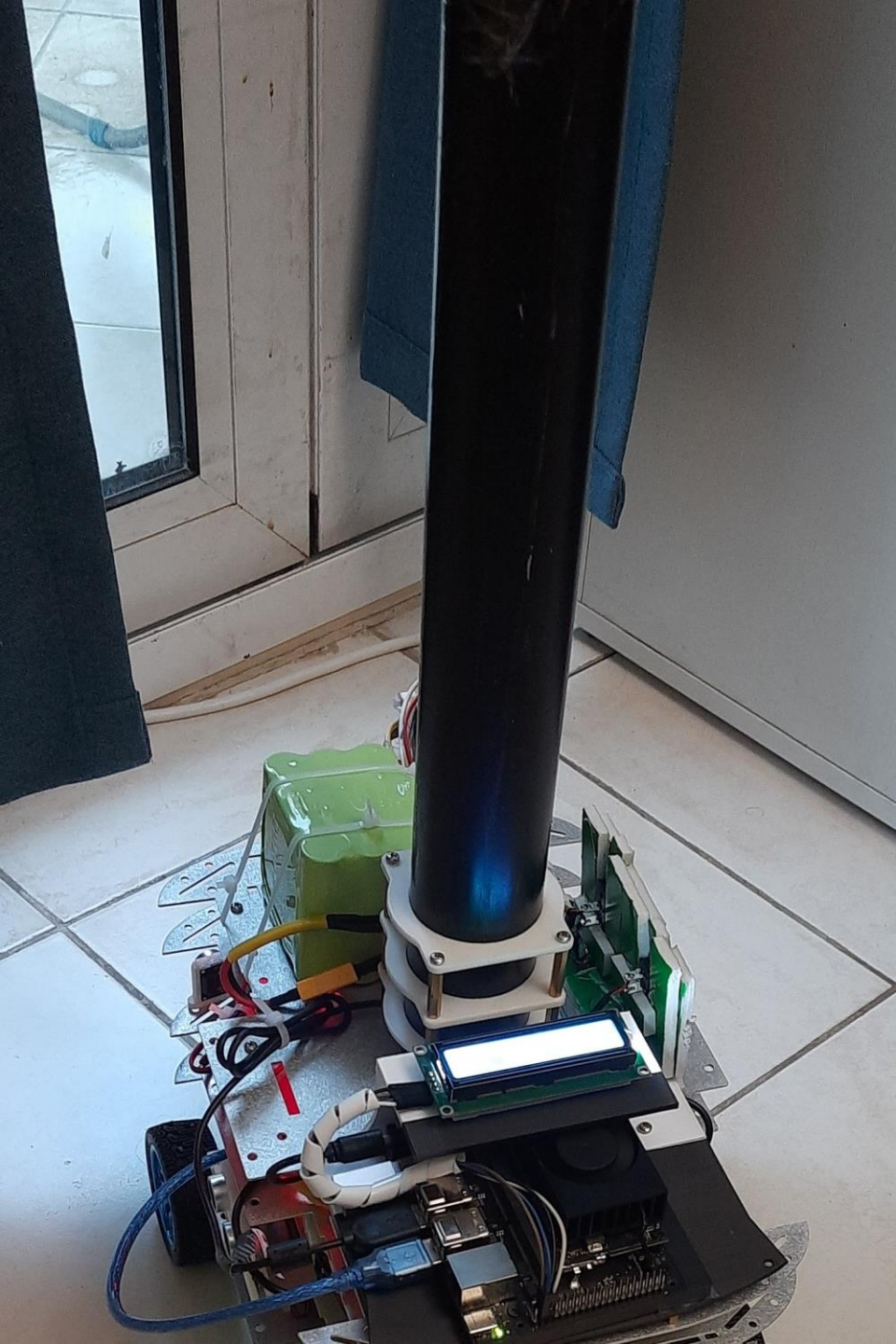


CARA – Companion Assistive Robot for the Aged

A low-cost GSM-based assistive robot for elderly care in developing countries

PDE4445 Robotics Dissertation Project

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MOTIVATION

The Care Gap in Developing Countries

A Growing Crisis

Rapidly ageing populations with insufficient infrastructure — many elderly live alone without consistent care.

Existing Solutions Fall Short

PARO, Pepper, and Care-O-Bot cost thousands of dollars and require Wi-Fi — neither affordable nor deployable in low-resource settings.

The Design Imperative

A sub-\$500, GSM-connected robot with onboard AI — viable where current systems are not.

Five Research Questions

1

Meaningful Assistive Support

Can a low-cost robot provide meaningful assistive support?

2

Onboard Vision for Fall Detection

Can low-cost onboard vision achieve reliable fall detection?

3

GSM Connectivity

Can GSM replace Wi-Fi-dependent connectivity for safety alerts?

4

Cost-Driven Trade-offs

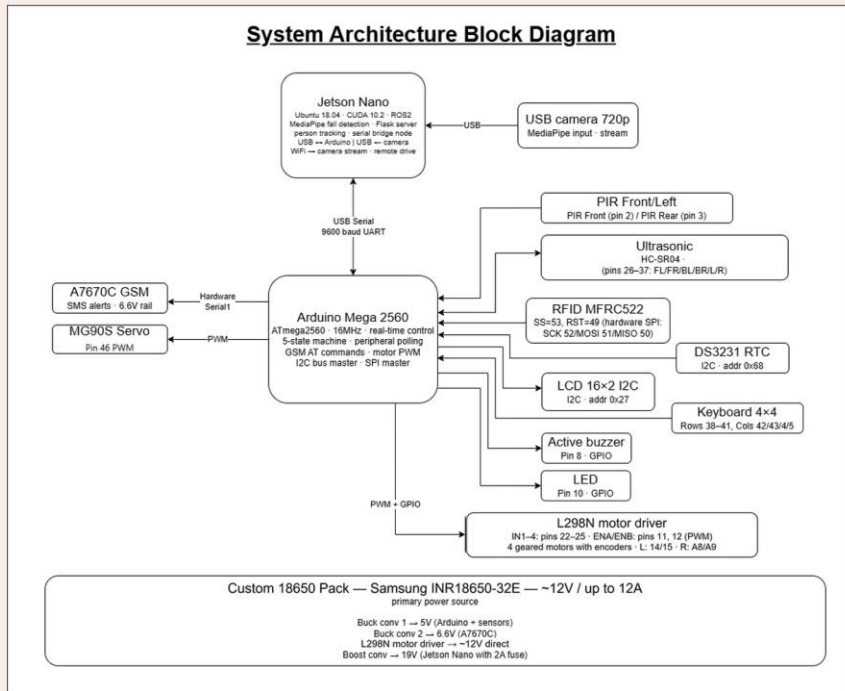
What trade-offs does a strict cost ceiling force?

5

Multimodal HRI at Low Cost

How can multimodal HRI be studied at this cost point?

Dual-Processor Architecture



Serial-linked dual-processor design separates safety-critical control from AI perception.

- **Arduino Mega 2560** — real-time motor control, obstacle avoidance, GSM alerts
- **NVIDIA Jetson Nano** — MediaPipe fall detection, person tracking, face recognition
- **GSM (LTE)** — all safety-critical alerts; no Wi-Fi dependency
- **Serial link** — inter-processor communication
- **Total BOM cost** — under \$500

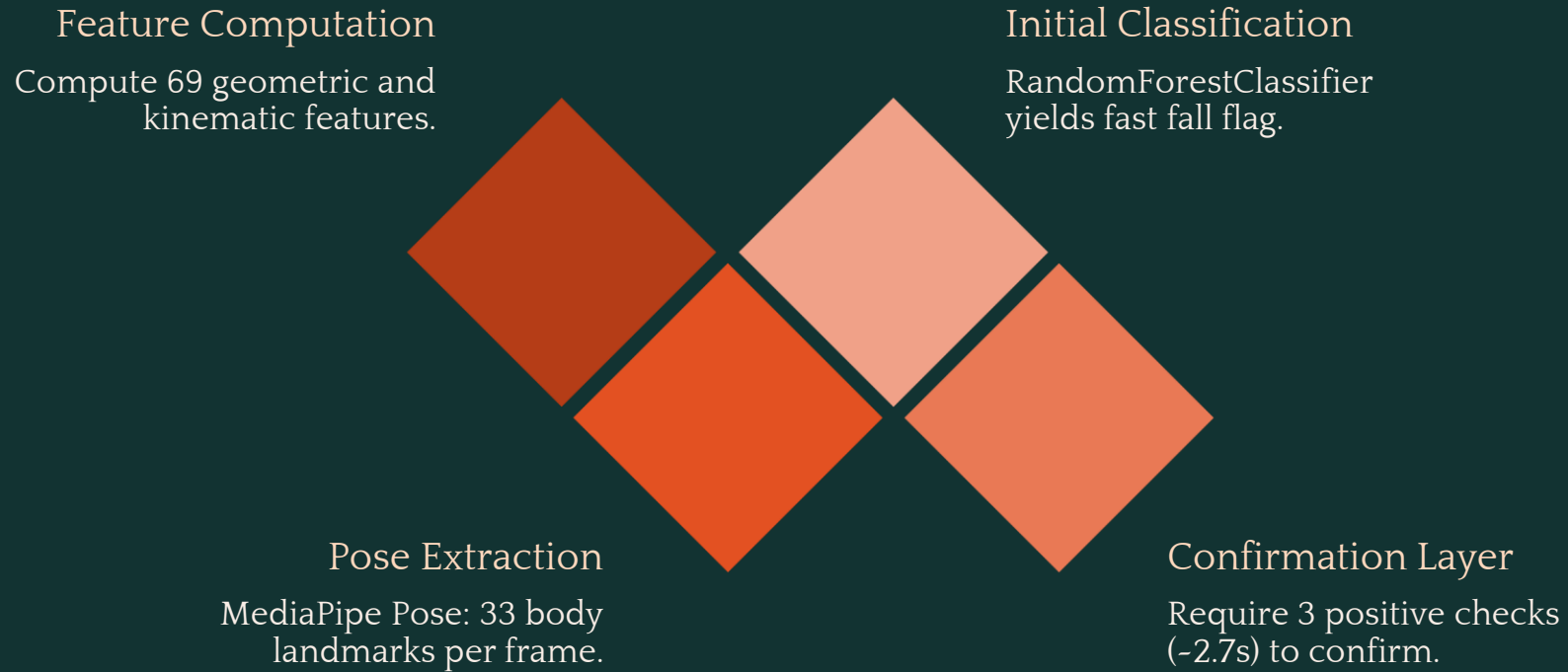


IMPLEMENTED FEATURES

Fifteen Implemented Features

- Autonomous navigation with obstacle avoidance
- AI-based fall detection (MediaPipe + RandomForest)
- GSM SMS alerts with delivery confirmation
- Medication reminders via RFID tag detection
- Live camera streaming
- Remote drive via web dashboard
- Companion greetings and face-recognition enrolment
- Colour-band tracking fallback mode
- Person tracking
- Encoder-based closed-loop speed control
- Dual-motor feedback drive system
- Inbound SMS logging
- Two-layer fall confirmation logic
- Software I2C recovery fallback
- Serial inter-processor communication protocol

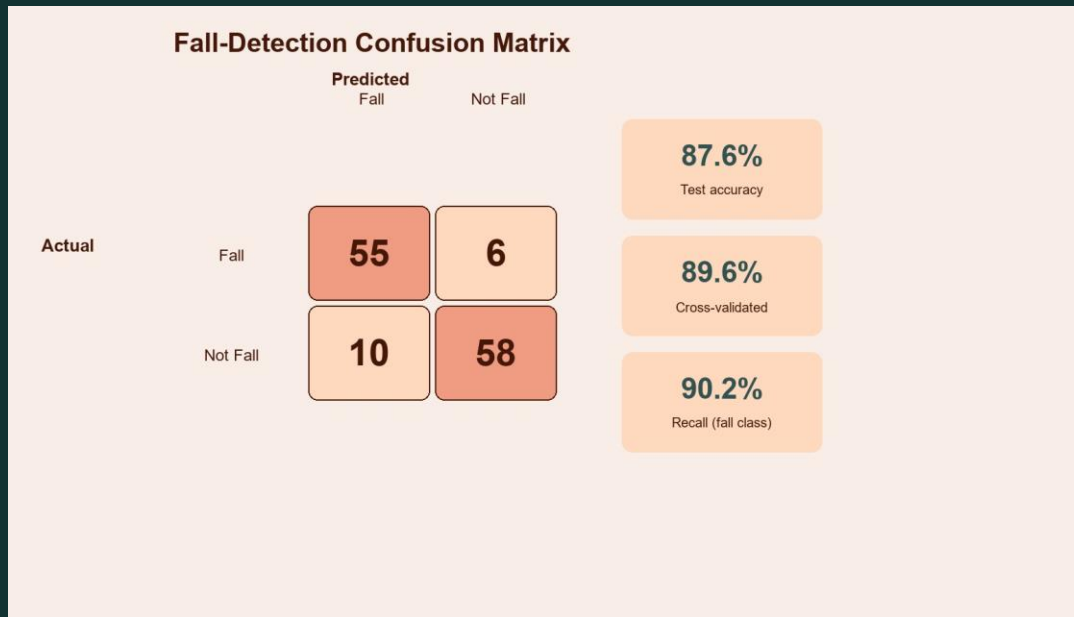
MediaPipe → Features → RandomForestClassifier



The two-layer confirmation — fast flag plus 3-consecutive-check verification over -2.7 s — reduces false positives from transient pose noise without compromising recall on genuine falls.

KEY RESULTS

Measured System Performance



Fall Detection

- **87.6%** test accuracy (target: 80%)
- **89.6%** cross-validated accuracy
- **90.2%** recall on fall class – prioritised over precision; a missed fall is worse than a false alarm

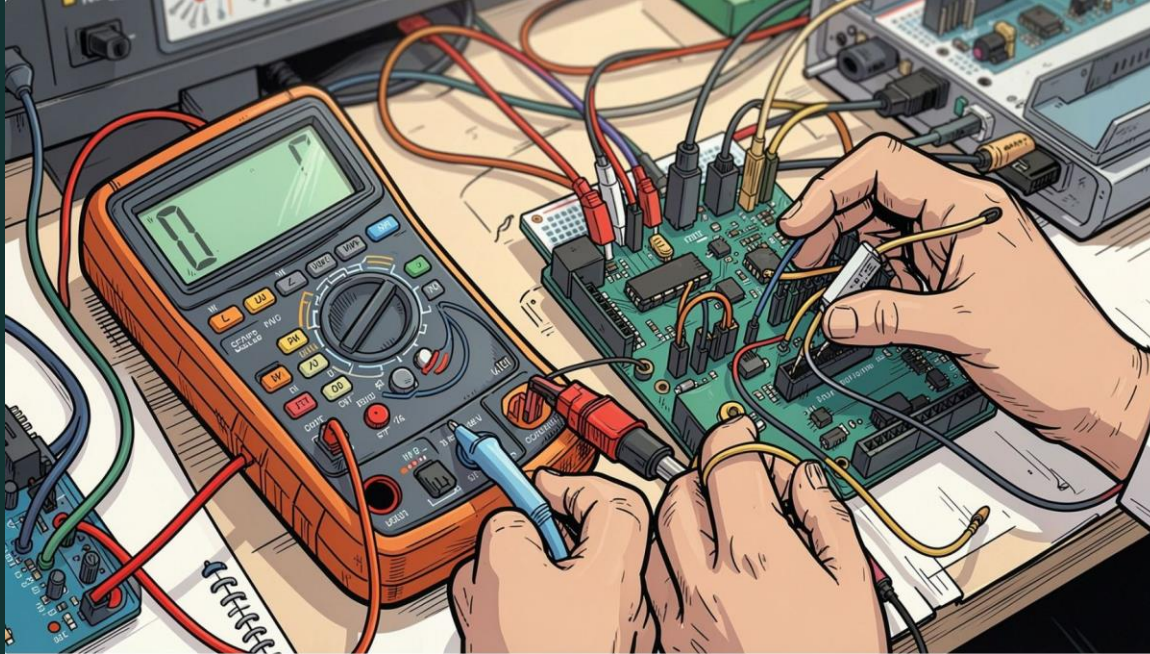
GSM Connectivity

- Full send, delivery confirmation, and inbound logging working end-to-end on replacement LTE hardware
- Original 2G module found network-incompatible – replaced during development

Motion Control

- 4-motor drive with dual encoder feedback; forward/reverse verified over repeated trials
- Obstacle-avoidance guard correctly halts the robot near obstacles

Systematic Fault Diagnosis as a Transferable Contribution



Two hardware faults were diagnosed and resolved using a repeatable elimination methodology — presented here as a transferable contribution, not merely a project-specific fix.

Fault 1 — Defective RFID Reader

Traced via power → pin → driver → wiring elimination; unit replaced.

Fault 2 — Damaged I2C Pin

Physically damaged pin on the Mega recovered by migrating to a software I2C bus on alternate GPIO pins.

LIMITATIONS

Honest Limitations

Single-Subject, Lab-Only Testing

All trials conducted with one participant in a controlled environment — real-world generalisation remains unvalidated.

In-Place Turning Fault

One drive side stalls under load during in-place turns; root cause under diagnosis.

Colour-Band Tracking Range

Reliable only at close range; tracking drops out beyond a few feet.

Multi-Person Face Recognition

Discrimination between multiple enrolled faces not yet formally tested.

Path to Commercial-Grade Reliability

01

Resolve Turning Fault

Diagnose and fix the asymmetric stall under load.

02

Formal Collision-Free Trial Count

Complete structured navigation trials with quantified success rates.

03

RTC-Based Medication Scheduling

Add real-time clock for autonomous, time-accurate reminders.

04

15-Feature End-to-End Trial

One continuous integrated demonstration of all features.

05

Ethical Clearance & Elderly Trials

Formal testing with elderly participants — the definitive validation step.

Thank You — Questions Welcome